Online Sales & Profitability

Problem Statement

- AdventureWorks currently lacks a comprehensive, real-time view of its online sales performance across different regions and product categories. This gap in visibility forces managers to rely on broken or fragmented data, making it difficult to identify rising market trends, optimize inventory for high-volume products or effectively allocate marketing budgets, ultimately hindering data-driven decision-making and profitability growth.

Value Hypothesis

- By implementing the "AdventureWorks Online Sales & Profitability Dashboard" the organization expects to significantly improve operational efficiency and strategic agility. Specifically, identifying high-volume but lower revenue categories will allow for targeted cross-selling campaigns to increase the average products per order. Furthermore, pinpointing top-performing regions will enable the replication of successful strategies in underperforming areas, potentially increasing overall online sales or revenue and optimizing supply chain cost.

Key Business Questions

- What is the total online revenue generated across all regions and product categories and how does it break down by geographic segment?
- Which regions and continue are driving the highest sales performance, and where should we allocate marketing budgets for expansion?
- What is the profit margin percentage by product category, and which categories offer best profitability versus volume trade-offs?
- How many physical units are sold per category, and how does this inform inventory management and demand forecasting?
- What is the current daily sales run rate , and are we on track to meet monthly and quarterly revenue or sales targets?
- What is the average order value and average products per order, indicating the effectiveness of cross-selling and upselling strategies?
- How many orders does the average customer place, and what does this reveal about customer loyalty and retention?
- What are the seasonal sales trends over time, and when should we prepare for peak demand periods?

Success Measures

- **Performance & Speed:** The dashboard must load and respond to all filter interactions in under 5-10 seconds ensuring a smooth user experience.
- **Data Accuracy & Trust:** All metrics must match official financial reports and the SQL data source with 100% flexibility.
- **Visual Clarity:** Executives must be able to understand the overall business health within seconds to minutes of viewing the dashboard.

- **Interactivity:** Users must be able to seamlessly drill down from high-level KPIs into granular, detailed data points without friction.

- **Actionability:** Every visual must drive specific, tangible business decisions rather than just displaying data or charts.

Stakeholders

- **CEO/Executive Teams:** Need a high-level performance overview to make strategic decisions regarding market positioning and resource allocation.

- **Regional Managers:** Need to compare regional performance to allocate resources effectively and replicate successful strategies across territories.

- **Sales Directors:** Need to identify sales trends over time and adjust marketing strategies based on product category performance and customer buying behavior.

- **Inventory Managers:** Need to forecast demand based on sales velocity and unit movement to optimize stock levels and warehouse allocation.

Data Requirements

- **Primary Data Source:** SQL Server - AdventureWorks Database 2025.
- **Fact Tables:** FactInternetSales
- **Dimension Tables:**
    - DimDate
    - DimSalesTerritory
    - DimProduct
    - DimProductCategory
    - DimProductSubcategory
- **Aggregated View:** Will be named as “view_MasterKPIs (pre-calculated view grouping data by Year, Month, Region, Country, and Category to ensure smooth dashboard performance).

Profitability & Promotion

Problem Statement

- AdventureWorks lacks a consolidated view of its sales promotions and its performance across online and reseller channels. This problem makes it difficult for managers to determine which promotions actually perform well and which do not, and which should be continued, adjusted, or discontinued.

Value Hypothesis

- By implementing the "AdventureWorks Promotions & Profitability Dashboard," the organization expects to improve promotion-related decisions by providing a clear view of its sales, profit, profit margin, and discount rate for the promotions it has run. This will help managers identify successful promotions, reassess underperforming promotions, and review promotional strategies.

Key Business Questions

- Which promotions generate the highest sales and profits?
- Which promotions should be reassessed or continued?
- Are there promotions where the discount rates are high but the profitability is low?
- How does the level of discounting relate to profitability across promotions?

Success Measures

- **Actionability**: Insights support the company's decisions to either continue or reassess promotions
- **Visual Clarity**: Users identify the strong and weak promotions quickly.
- **Data Accuracy**: Sales, profit, margin, and discount metrics are accurate and reflect the source data.
- **Interactivity**: Users can select a promotion and examine its performance.

Stakeholders

- **Sales Managers**: Evaluate promotion performance and decide on possible next steps.
- **Marketing Managers**: Assess the effectiveness of discount strategies.
- **Business Executives**: Monitor overall promotional profitability and resource allocation.

Data Requirements

- Primary Data Source: SQL Server – AdventureWorksDW2025
- Tables used:
    - **Fact Tables:**
        - FactInternetSales
        - FactResellerSales
    - **Dimensions Tables:**
        - DimPromotion
        - DimProduct
        - DimProductCategory
        - DimProductSubcategory
    - **Supporting Table:**
        - Promotion Selector

Customer Segmentation

Problem Statement

- AdventureWorks lacks prior knowledge on purchasing behavior from their customers as first-time buyers may not be impressed by the available products or prices put by the company, making revenue grow at a slow pace. Without establishing such marketing policies and strategies, customer retention can be diminished, causing AdventureWorks to close their business.

Value Hypothesis

- By implementing the "AdventureWorks Customer Segmentation Dashboard", the organization expects to maintain or improve the customer retention to generate more revenue with the help of understanding customer demographics such as career, age, yearly income, gender, and family structure. Marketing strategists will be able to plan on opening for promotions, discounts, and loyalty programs to customers going back to their business.

Key Business Questions

- Who are the customers who purchased the most?
- What are the careers or jobs commonly taken by customers?
- How much do customers earn?
- How much money would customers spend buying products at AdventureWorks?
- How fast does the number of first-time buyers increase as time passes by?
- How significant is the income earned yearly by customers based on how many children they have?

Success Measures

- **Customer Retention -** Encourages first-time buyers to keep on buying with no hesitation
- **Customer Lifetime Value Growth -** Increase customer volume to generate more revenue
- **Customer Demographics -** Develop customer profiles to curate what products they will be needed
- **Revenue Optimization -** Understanding different income ranges from customers to match their budget goods and services
- **Effective Marketing and Promotion -** Create marketing campaigns based on the customer demographics.

Stakeholders

- Marketing Committee - Makes customer profiles and plans for strategies to maintain or improve customer retention rate.
- Sales Committee - Formulates sales policies and strategies to generate revenue in the company.
- Customer Relations Officers - Ensures that customers will be provided goods and services in a proper manner.

Data Requirements

- **Primary Data Source:** SQL Server - AdventureWorks Database 2025
- **Table Used:**
    - **Fact Tables:** FactInternetSales, FactSurveyResponse
    - **Dimensions Tables:**
        - DimCustomer
        - DimGeography